



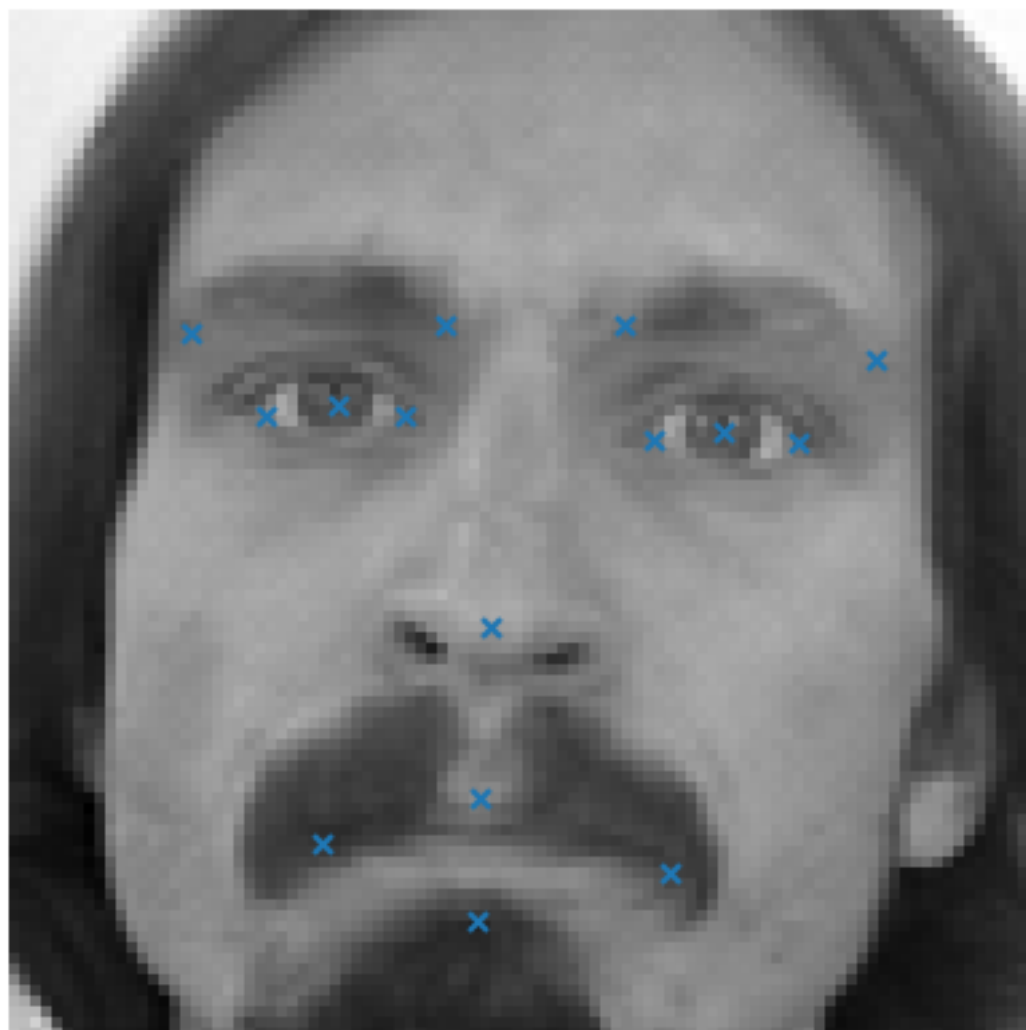
FACIAL KEYPOINTS DETECTION

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INTRODUCTION

Sample image with available facial keypoints



- Facial Keypoints Detection consists of locating points of a face
- Multi-output regression problem, we predict coordinates (x,y)

It is a problem where the model that solves it must understand facial geometry, not only classify an image.



PROJECT GOAL

BASELINE

For each keypoint, calculate the **mean of its coordinates** in the training set and always introduce that **same prediction for all images**.

GOAL

The goal is to study if models can actually **learn useful visual information** beyond the average face geometry and analyze **different complexities**.

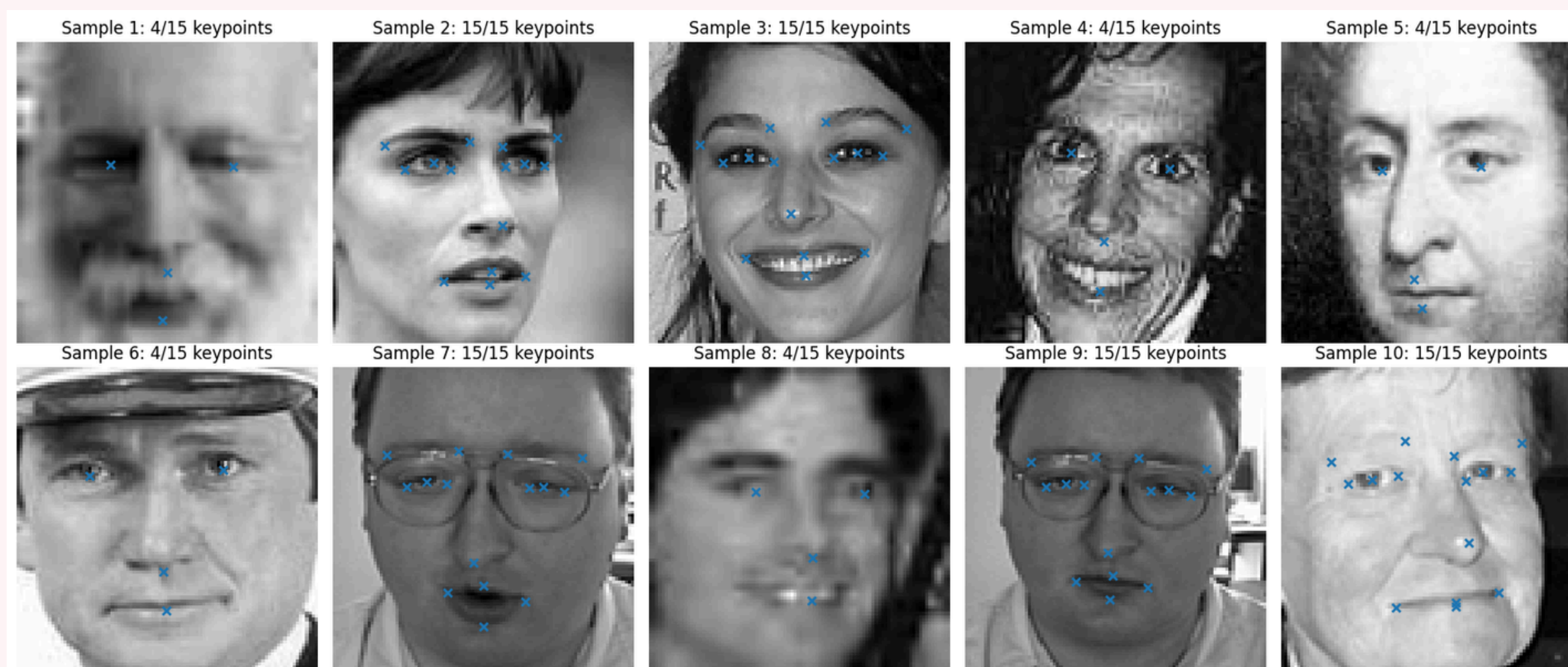
HOW

Comparing different models implemented by us to predict facial landmarks in images, using a fair and common evaluation.





DATASET



- **7049 grayscale images** of 96×96 (9216 pixels) with up to 15 facial keypoints, each with x,y coordinates.
- **Not all of them have 15 keypoints** (30% yes 70% no)
- They are mostly **frontal** images.





MISSING KEYPOINTS

- Eliminating incomplete samples would make us lose 70% of the dataset
- We create a **mask** to know which coordinates exist. The NaNs are substituted by **means** to keep the tensor size.

Thanks to the mask the loss and other metrics only consider valid coordinates

SPLITTING STRATEGY

- 80% train and 20% validation (**stratified**)
- The official Kaggle test set has no ground-truth labels, so we split the training portion again, reserving 15% of it as an **internal test set**.

Approximate final split of 68% train, 20% validation, and 12% internal test.

STATE OF THE ART



HEATMAPS

The model predicts a **probability map** for each landmark instead of coordinates directly.

ANCHORFACE

Uses predefined reference face shapes, called **anchors**, as starting points. Instead of predicting all landmarks from zero, the model compares the face with these anchors and **adjusts them to fit the specific image**.

- The dataset used is small, low-resolution, and mostly frontal. More advanced models like AnchorFace or heatmap-based transformers would have added a lot of **complexity** relative to the main objective.
- Our objective was to build a clear and controlled comparison with models implemented from scratch, aligned with the **content of the course**.

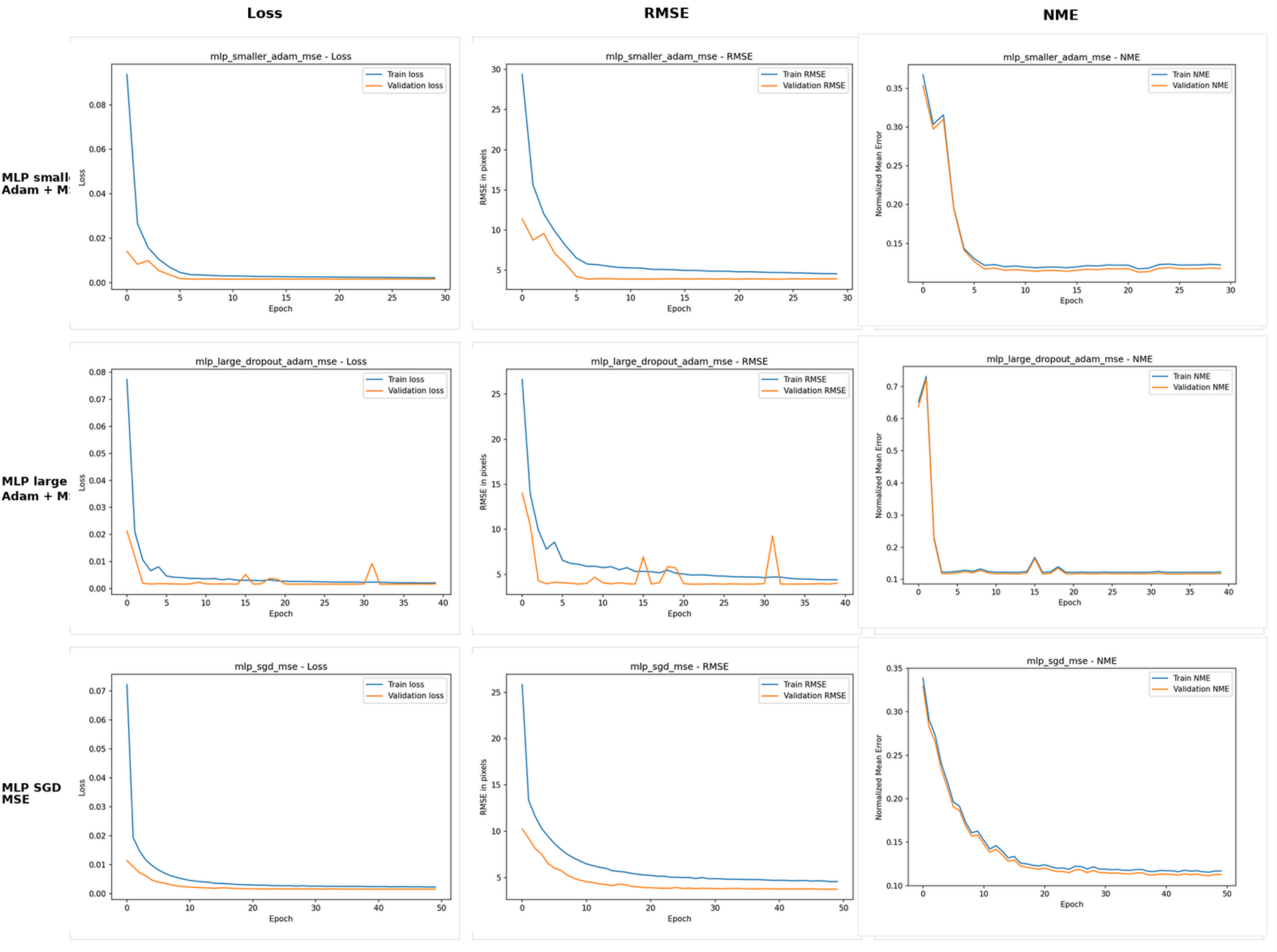


EVALUATION METRICS

- Loss
- RMSE - Main benchmark
- MAE - Average prediction error
- NME - Normalized error (inter-ocular distance)

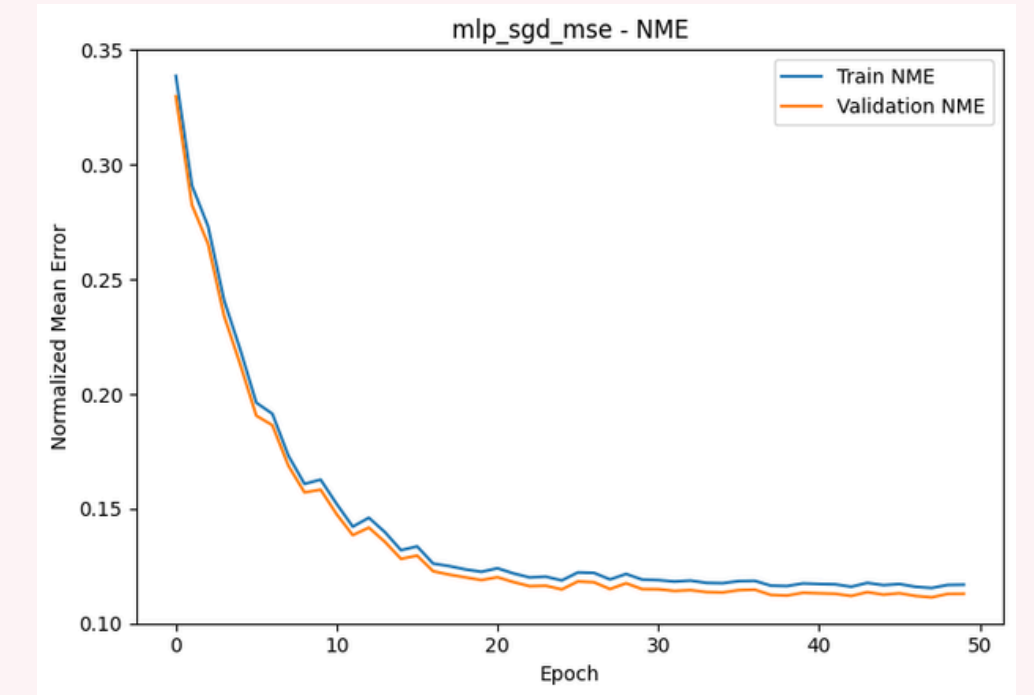
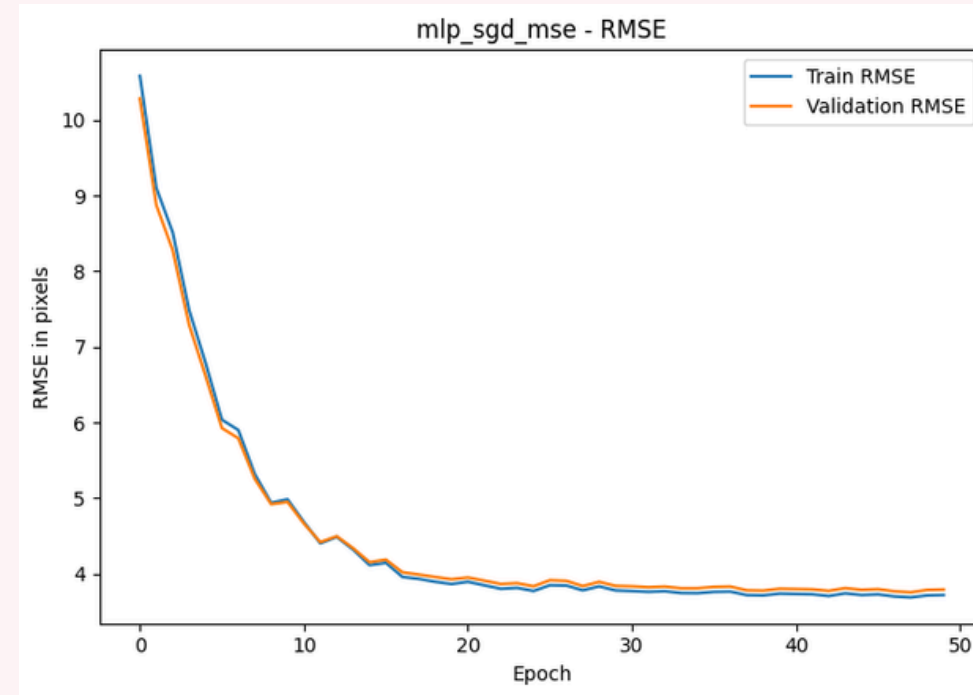
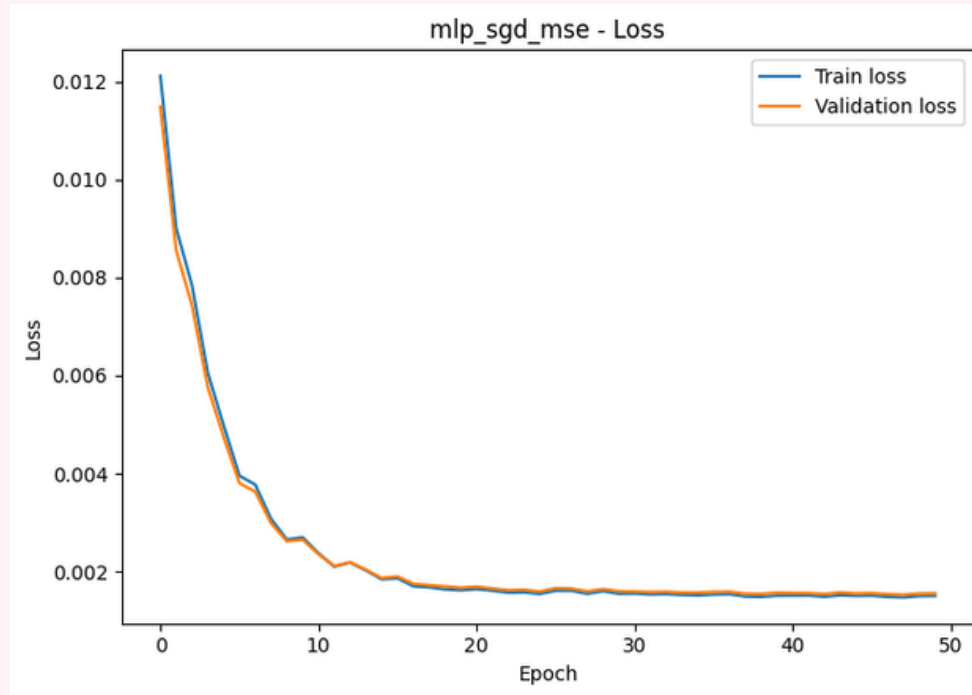
EXPERIMENT 1: MLP

	CONFIGURATION 1	CONFIGURATION 2	CONFIGURATION 3
Hidden Layers	512, 256, 128	512, 256, 128	1024, 512, 256
Optimizer	Adam	SGD	Adam
Dropout	0.3	0.3	0.5
Weight Decay	No	No	Yes
Focus	Baseline MLP	Optimizer Comparison	Larger + Regularized MLP



EXPERIMENT 1

BEST EXPERIMENT

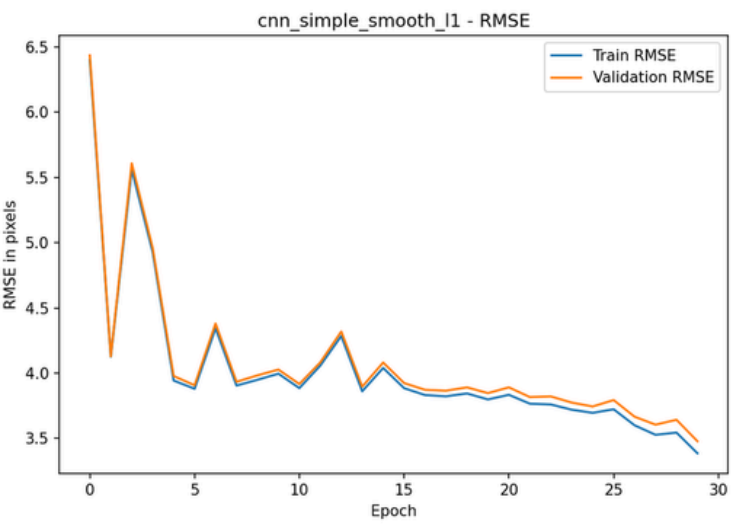
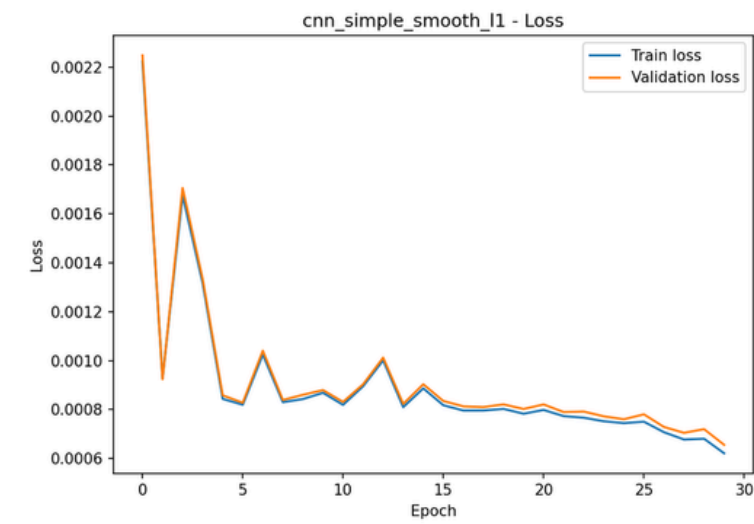


	CONFIGURATION 1	CONFIGURATION 2 - BEST	CONFIGURATION 3
RMSE	3.791	3.754	3.912
MAE	2.566	2.528	2.653
NME	0.113	0.111	0.117
Focus	Smaller MLP	SGD	Large MLP

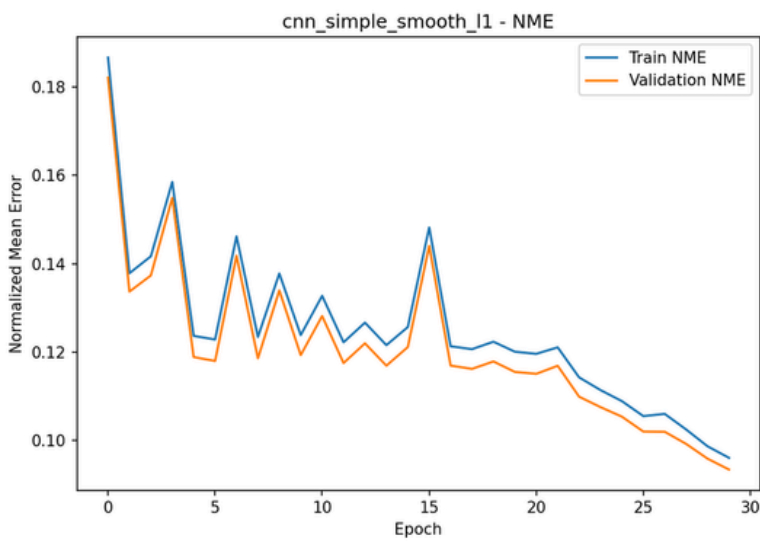
EXPERIMENT 2: Simple CNN

	CONFIGURATION 1	CONFIGURATION 2	CONFIGURATION 3
Conv Channels	32,64,128	16,32,64	32,64,128
Fully Connected layers	256	256	256
Loss	MSE	MSE	Smooth L1
Optimizer	Adam	Adam	Adam
Focus	Baseline CNN	Capacity Study	Loss Comparison

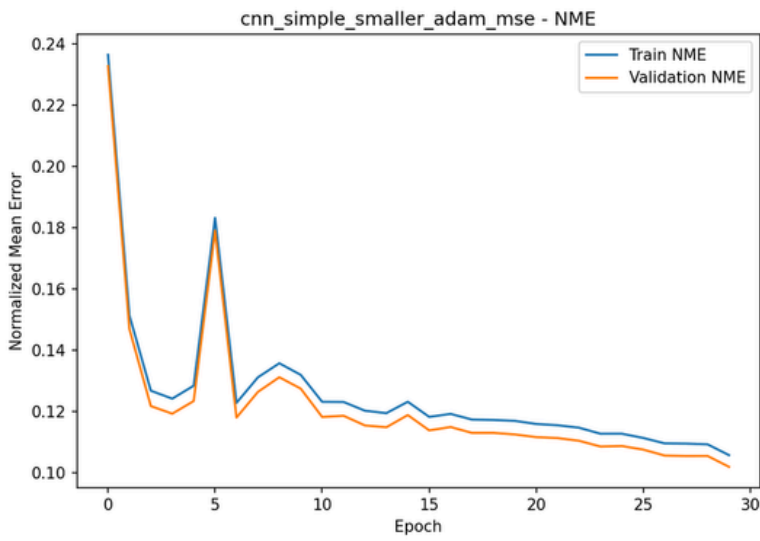
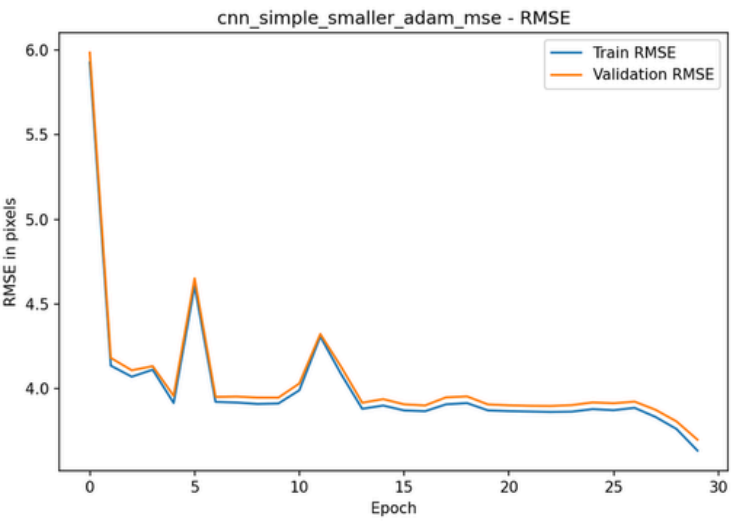
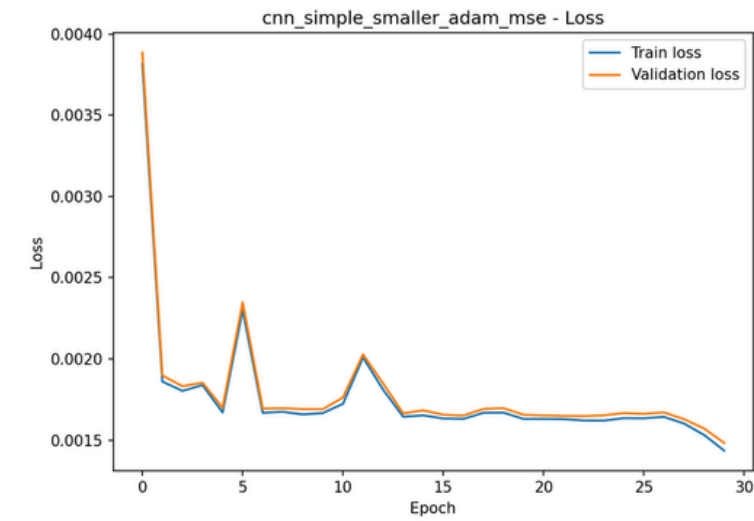
cnn_simple_smooth_l1



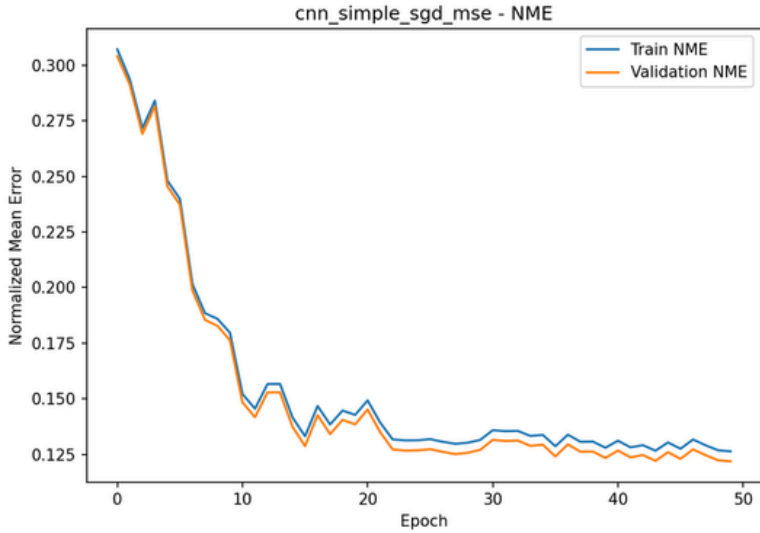
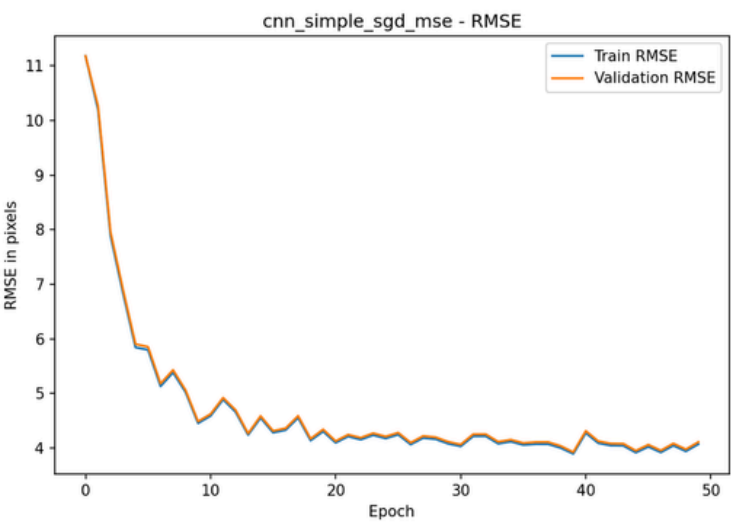
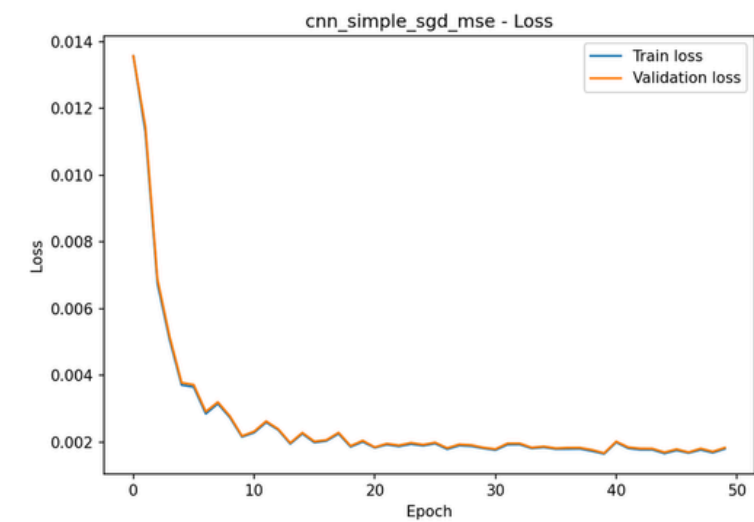
NME



cnn_simple_smaller_adam_mse

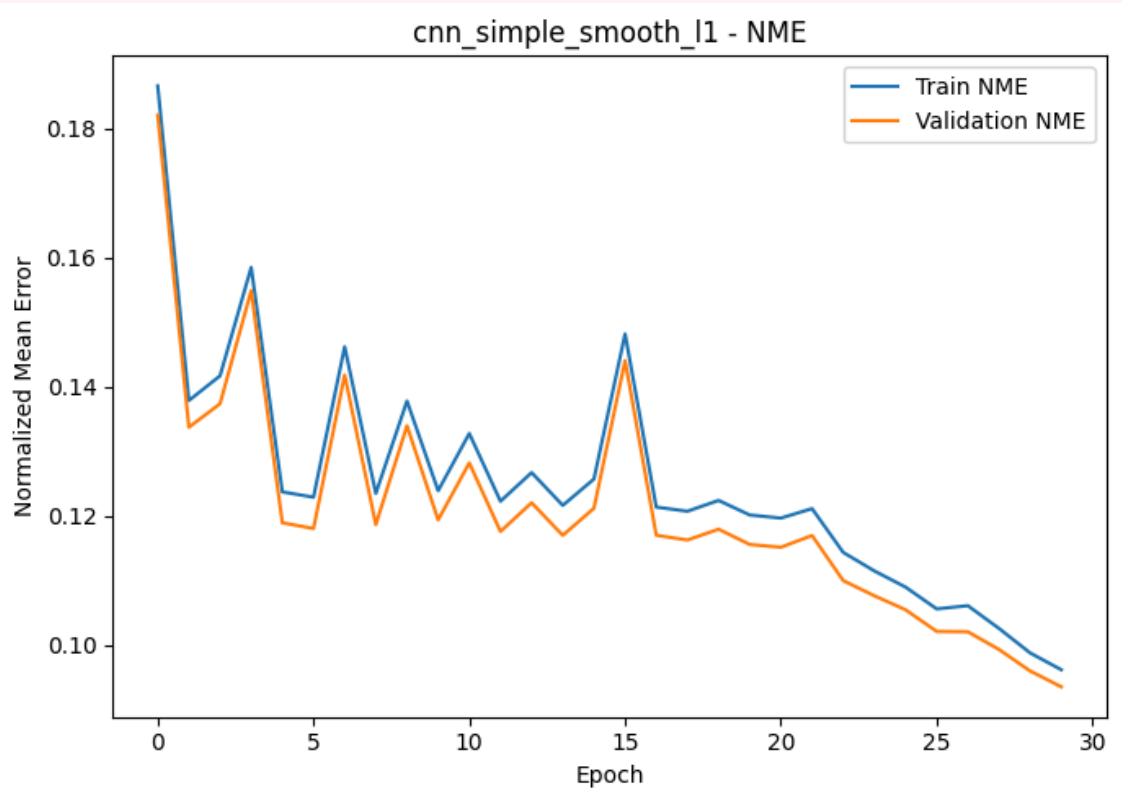
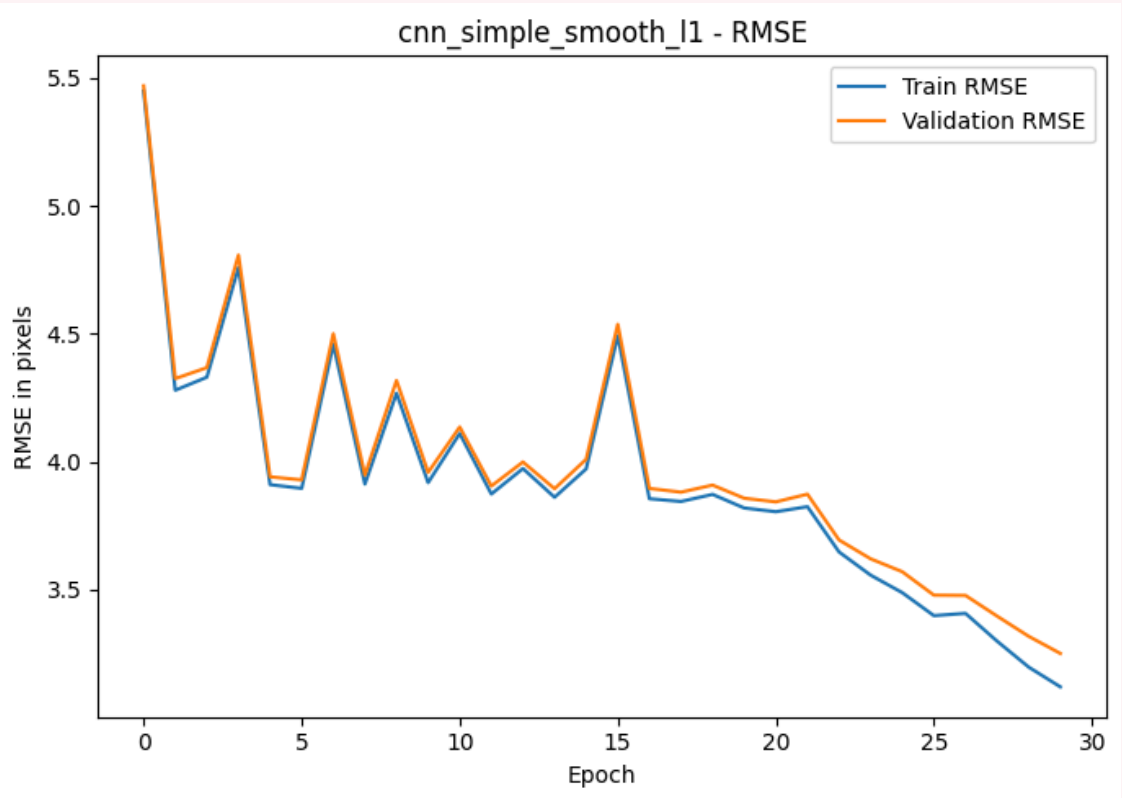
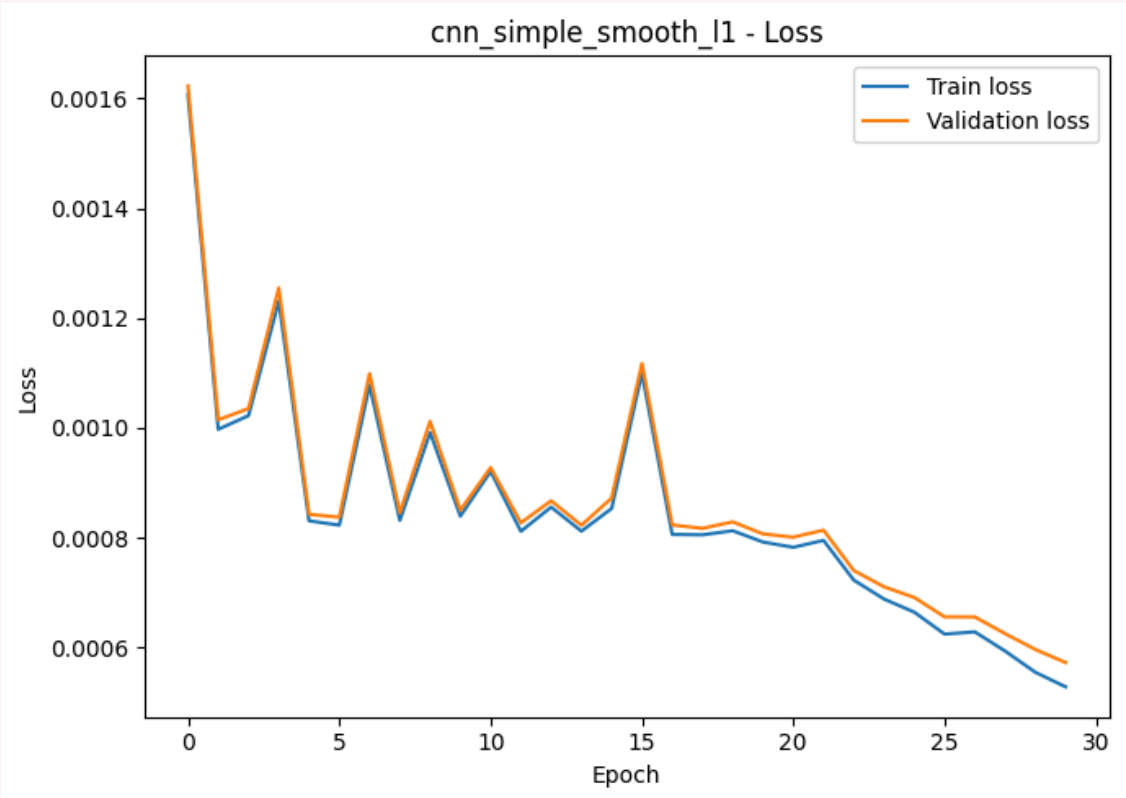


cnn_simple_sgd_mse



EXPERIMENT 2

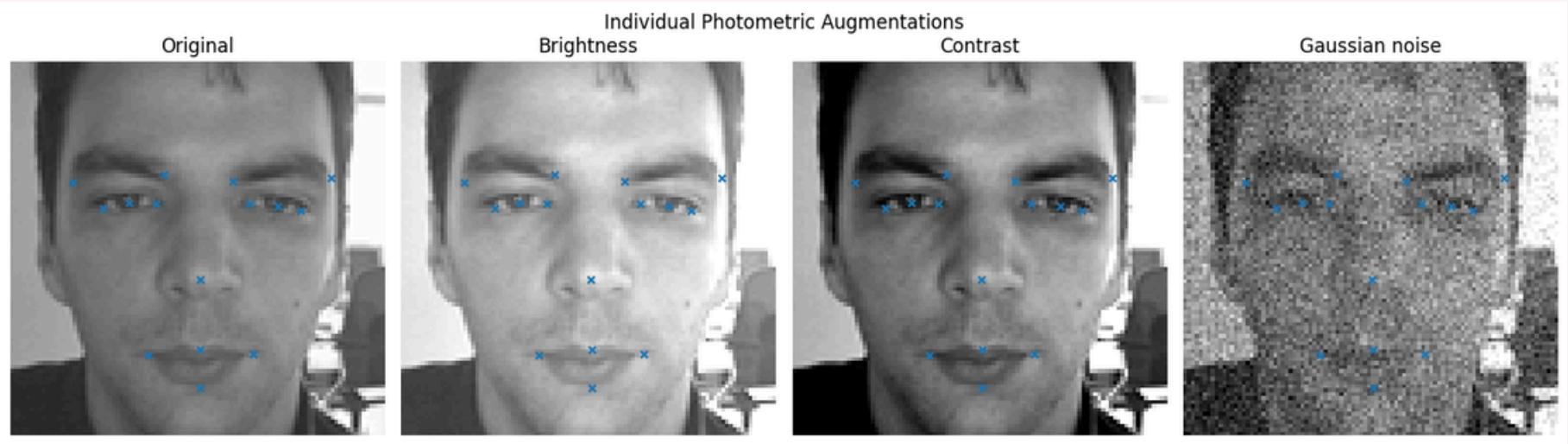
BEST EXPERIMENT

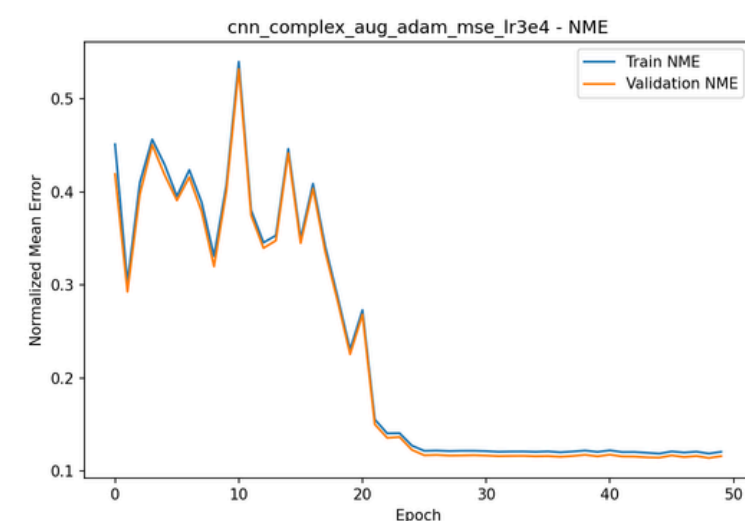
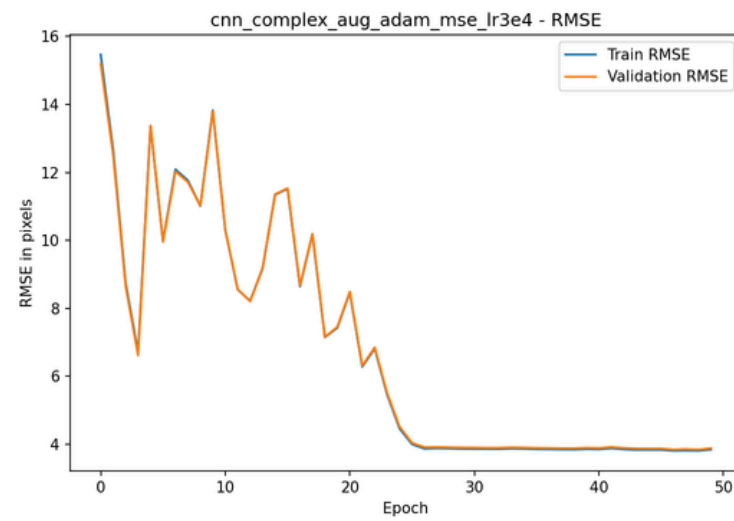
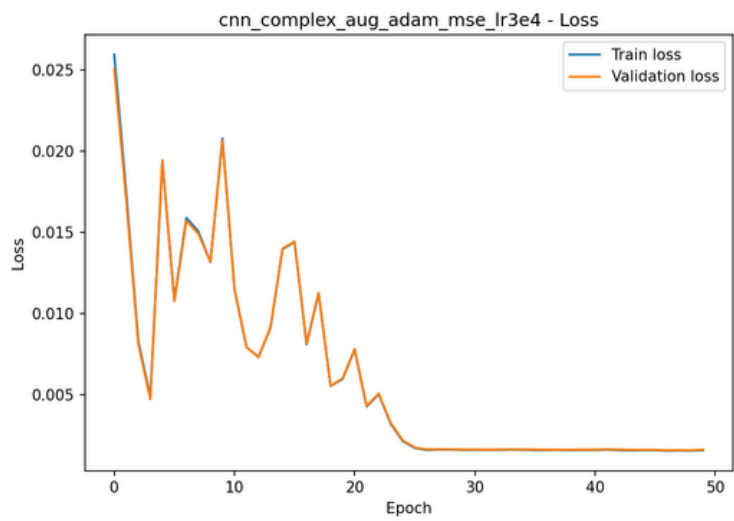
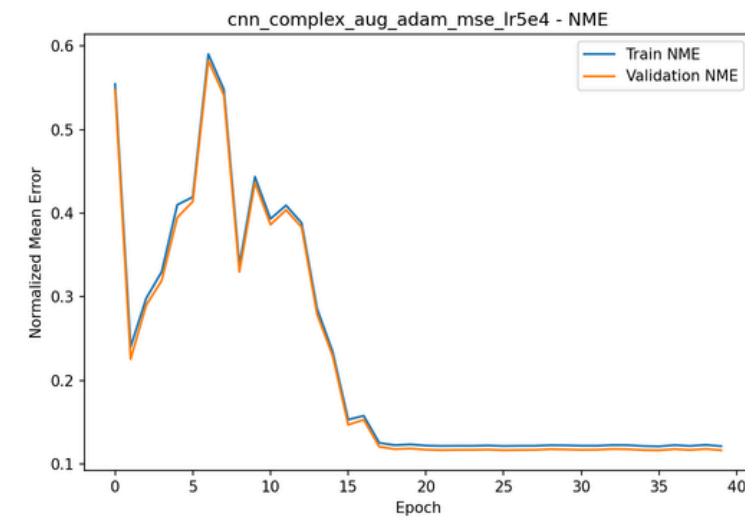
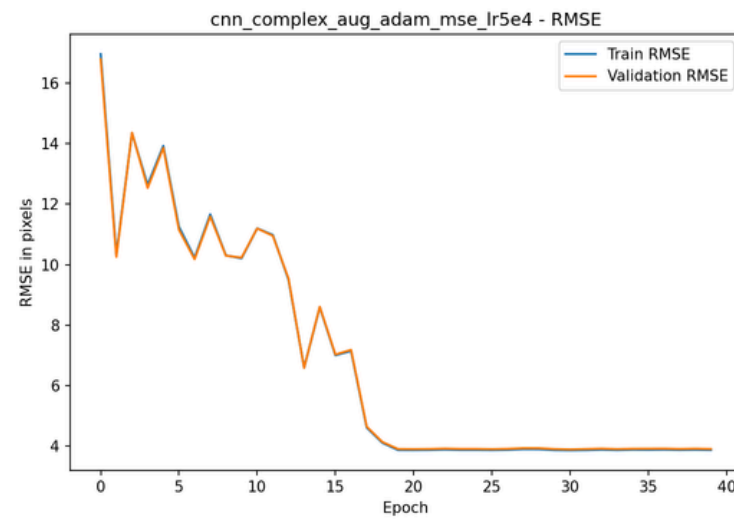
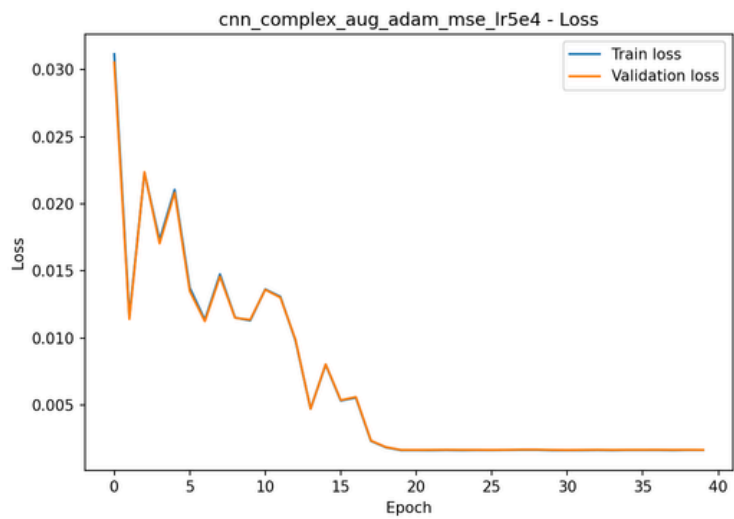
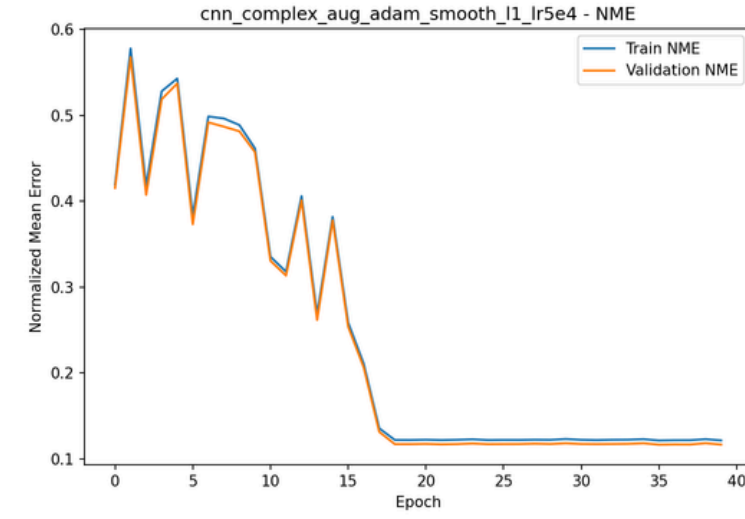
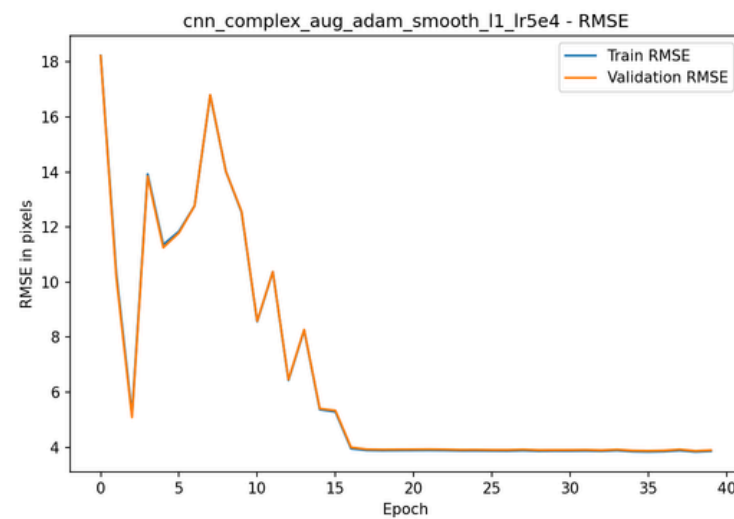
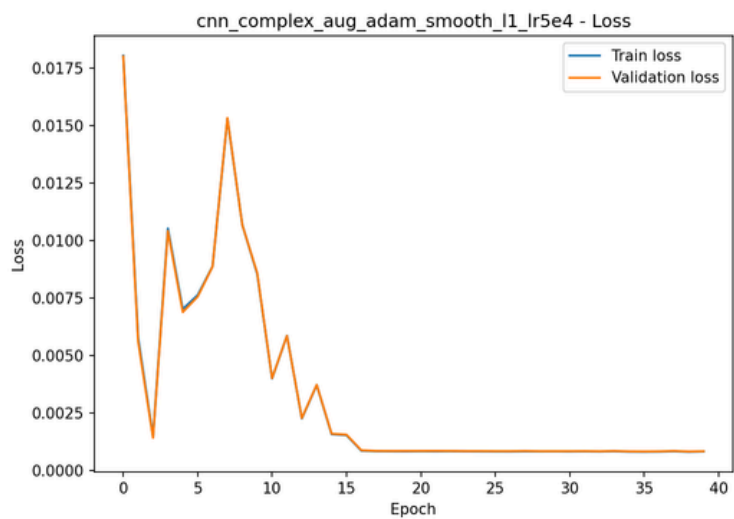


	CONFIGURATION 1	CONFIGURATION 2	CONFIGURATION 3 -BEST
RMSE	3.774	3.482	3.250
MAE	2.561	2.310	2.124
NME	0.113	0.102	0.093
Focus	Baseline CNN	Capacity Study	Loss Comparison

EXPERIMENT 3

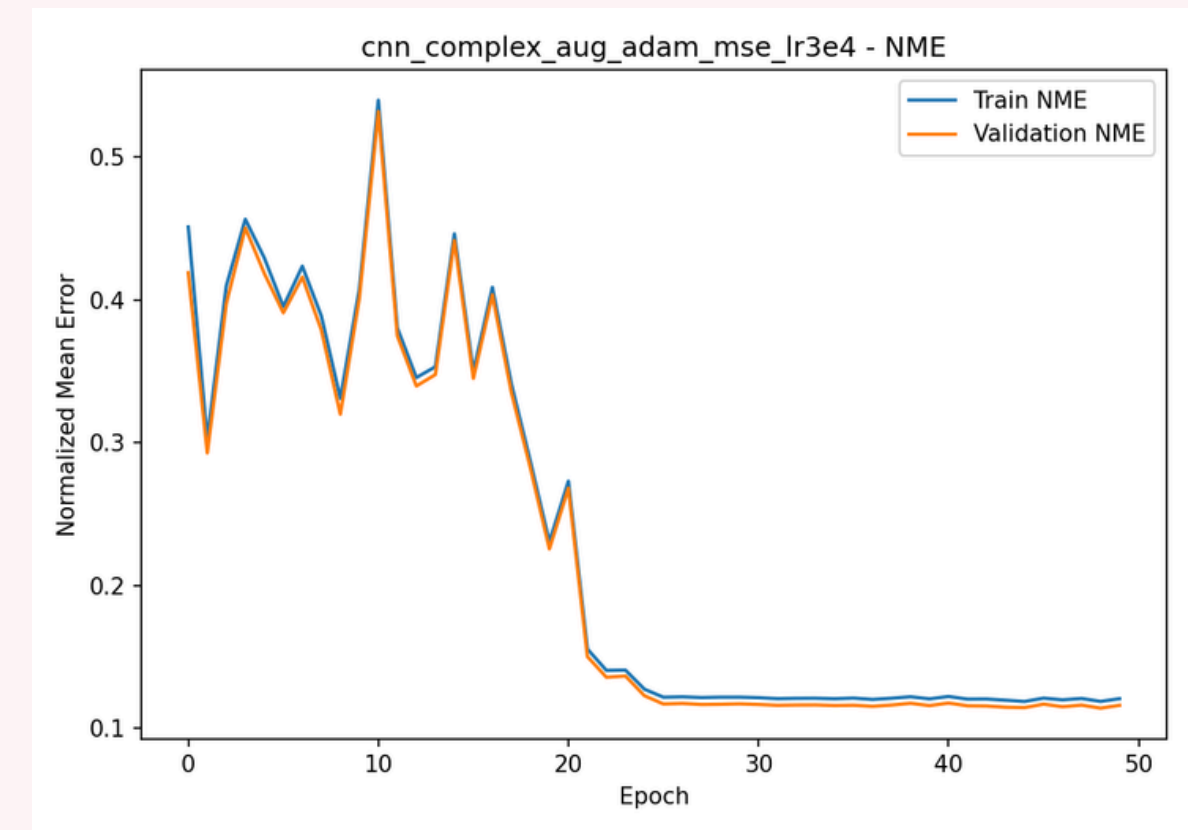
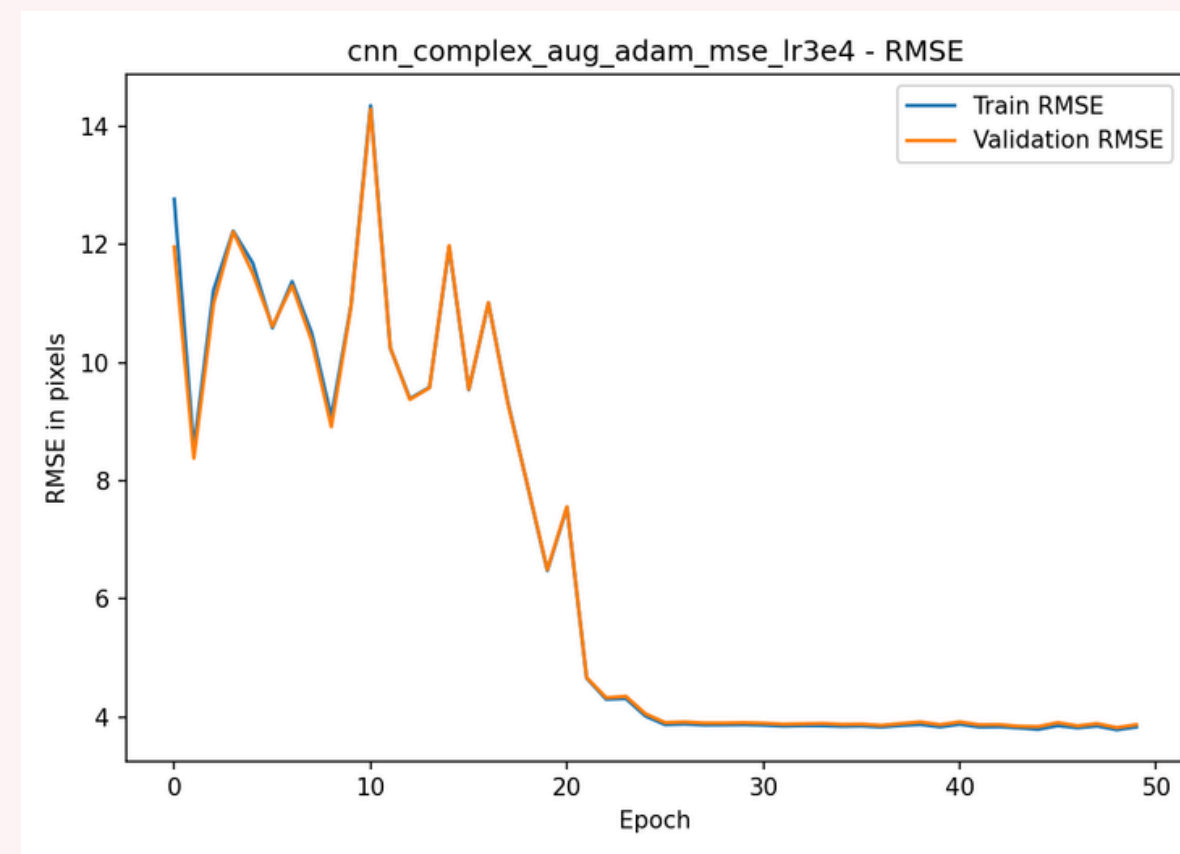
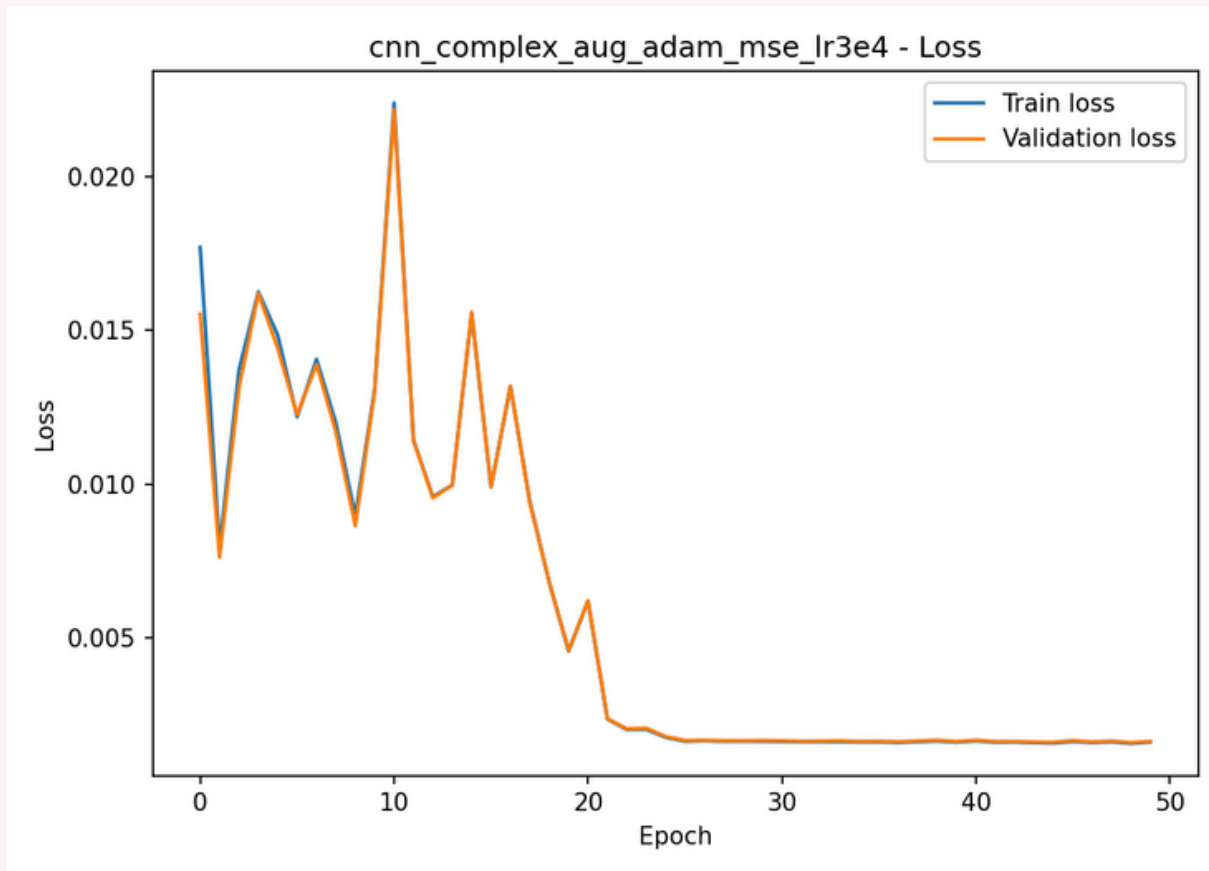
	EXP 2	EXP 3
Conv channels	32, 64, 128	32, 64, 128, 256
Fully Connected layers	256	256, 512
BatchNorm	NO	YES





EXPERIMENT 3

BEST EXPERIMENT



Baseline vs CNN Complex Results

Experiment	Val RMSE	Val NME	Test RMSE	Test NME
mean_keypoint_baseline	3.9078	0.1169		
cnn_complex_aug_adam_mse_lr3e4	3.8218	0.1140	3.7424	0.1166



EXPERIMENT 4

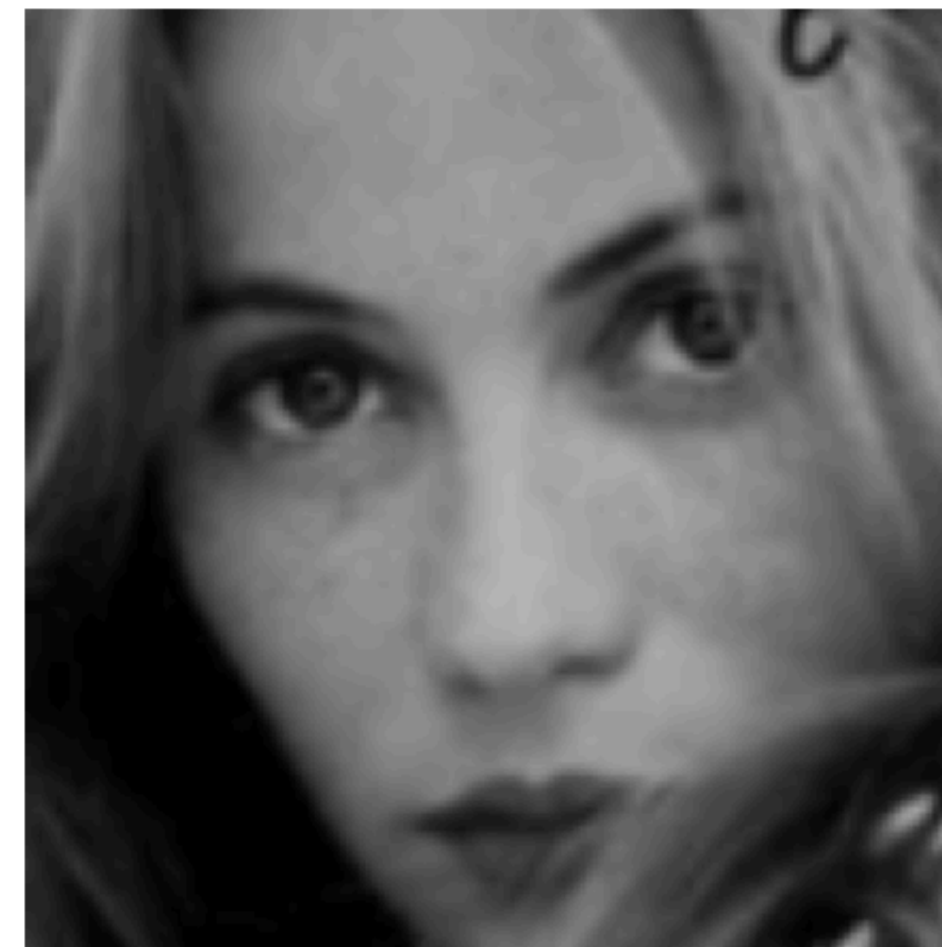
Youtube Videos Facial Keypoints (68 landmarks 3D with RGB colors)

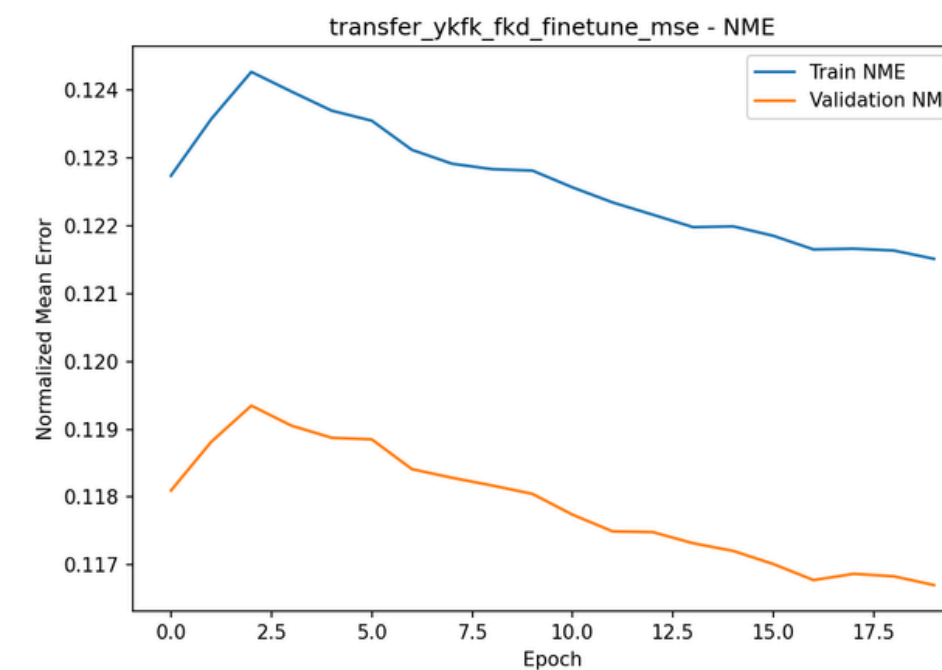
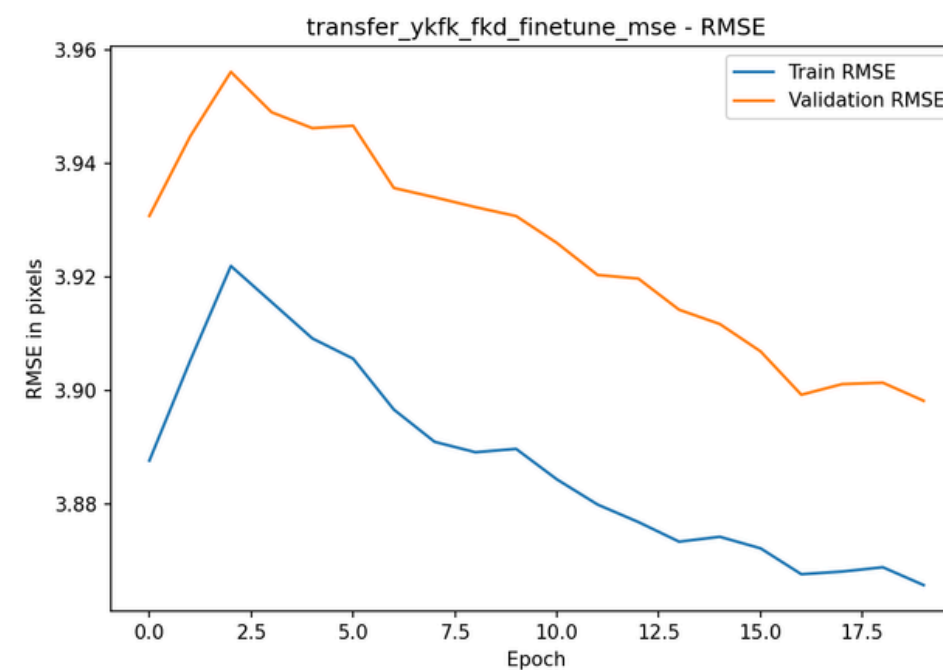
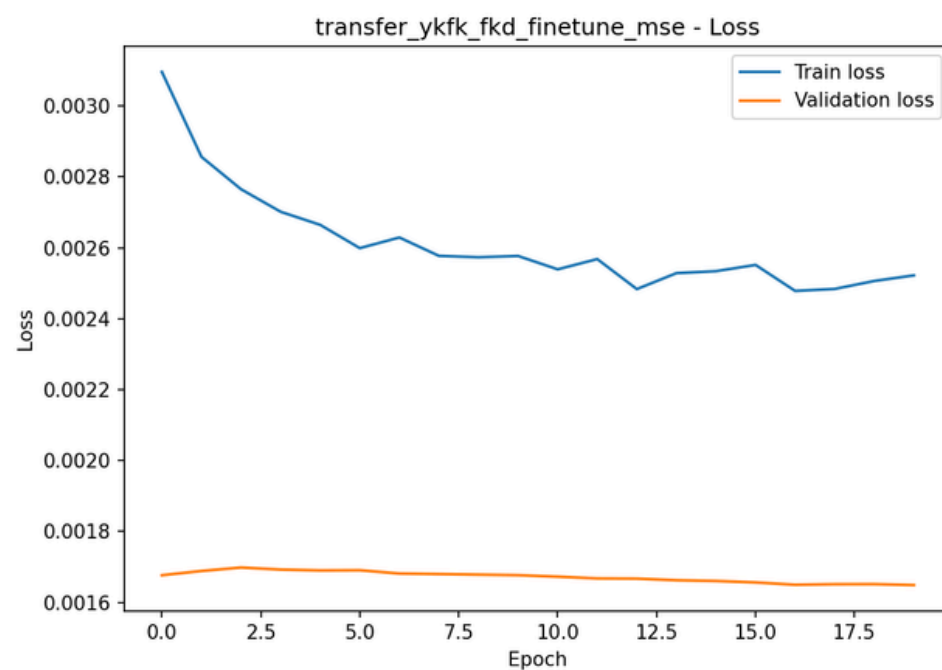
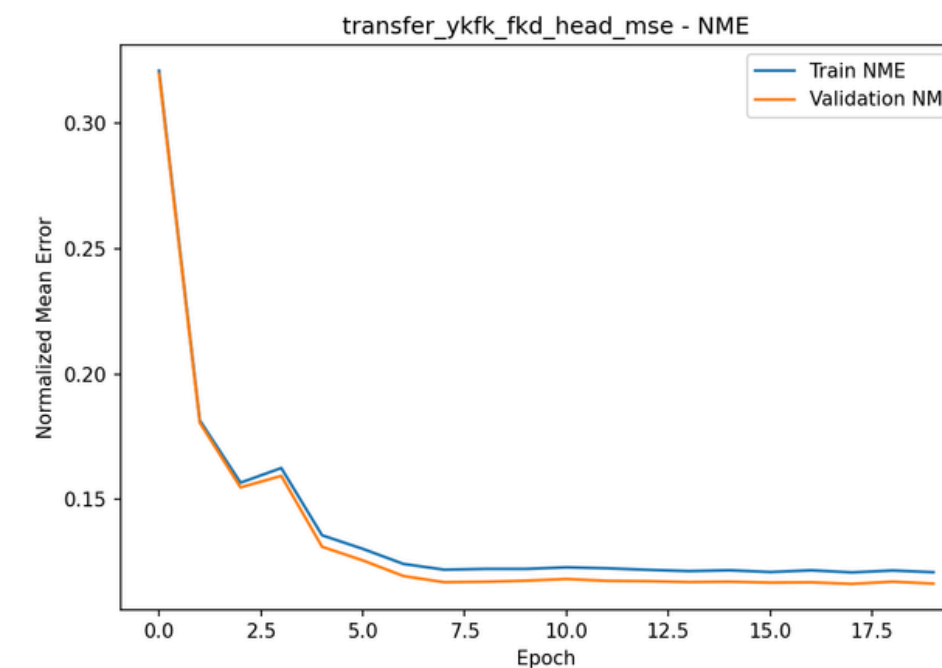
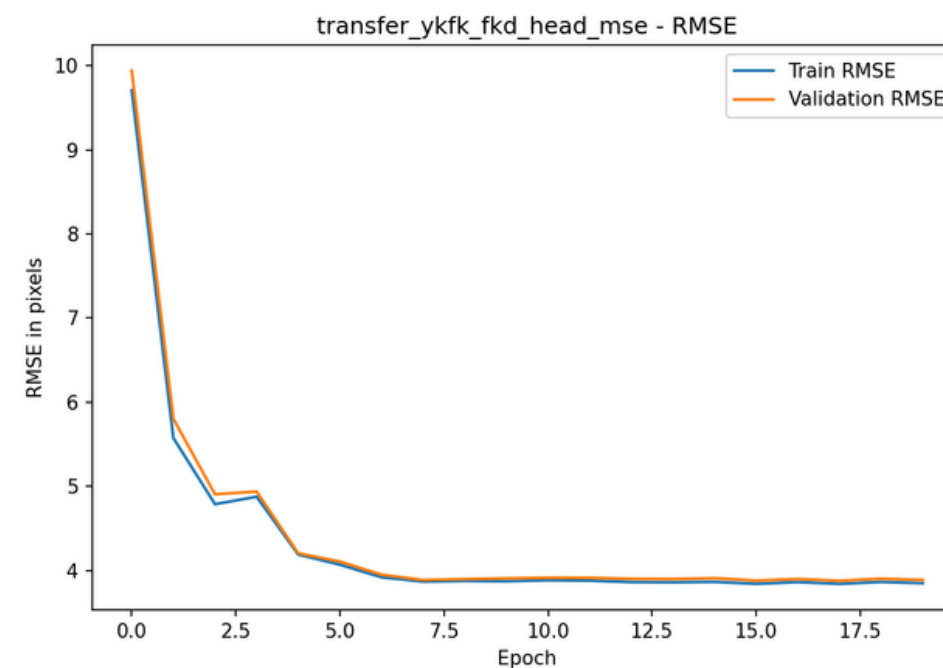
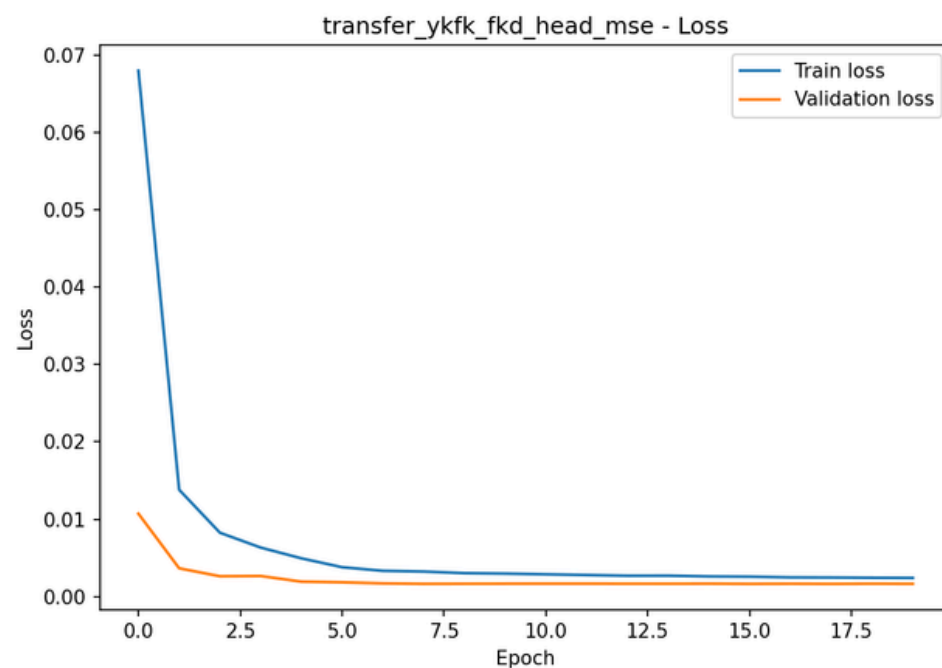
MobileNetV2 + ImageNet + Fully Connected Layer (68,3)

Original Image (96x96, 1 channel)



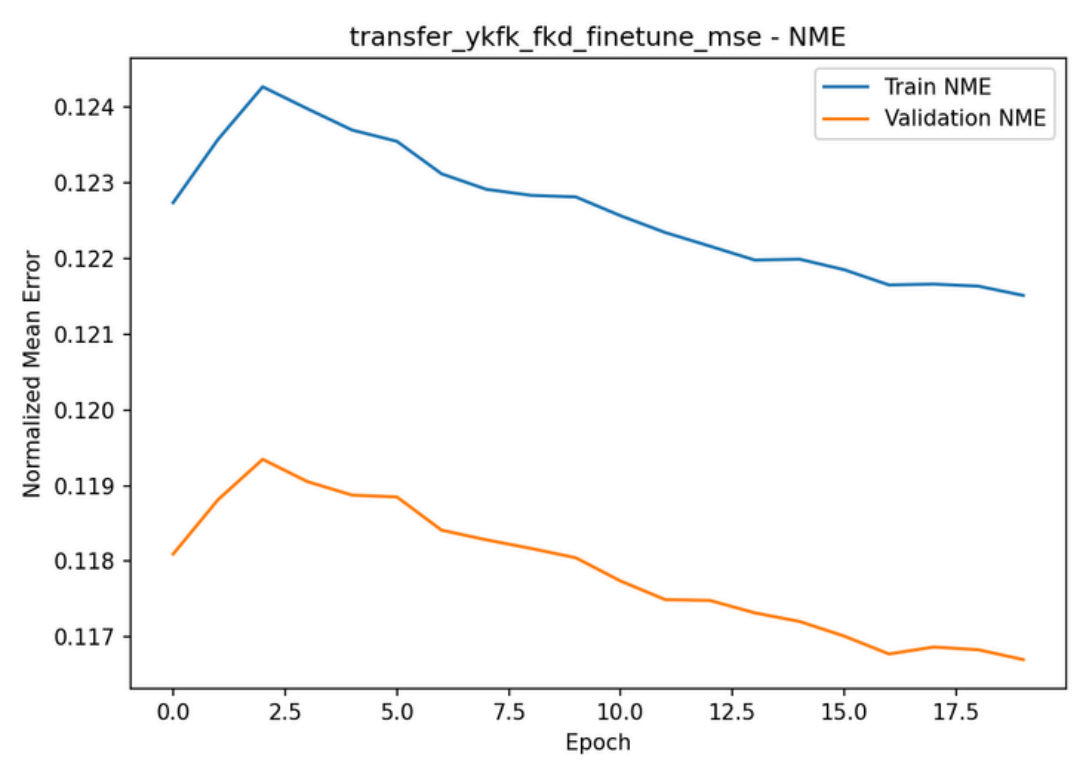
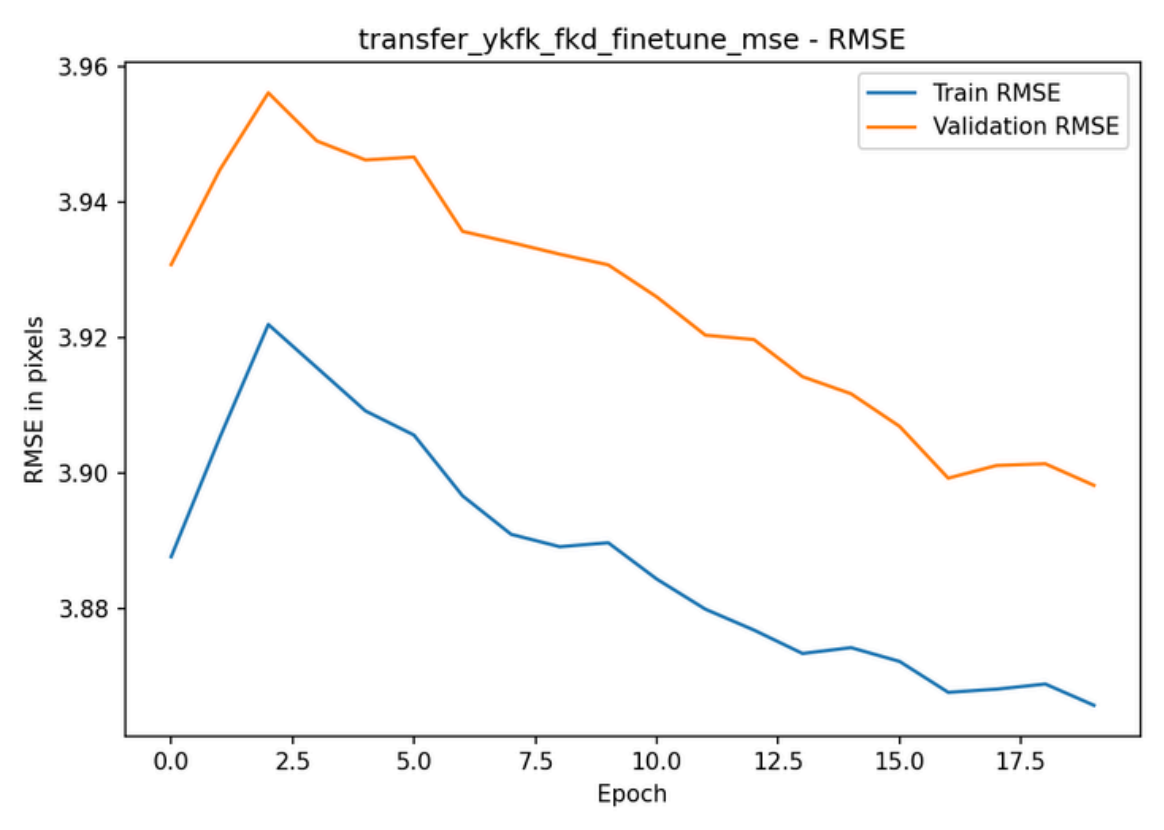
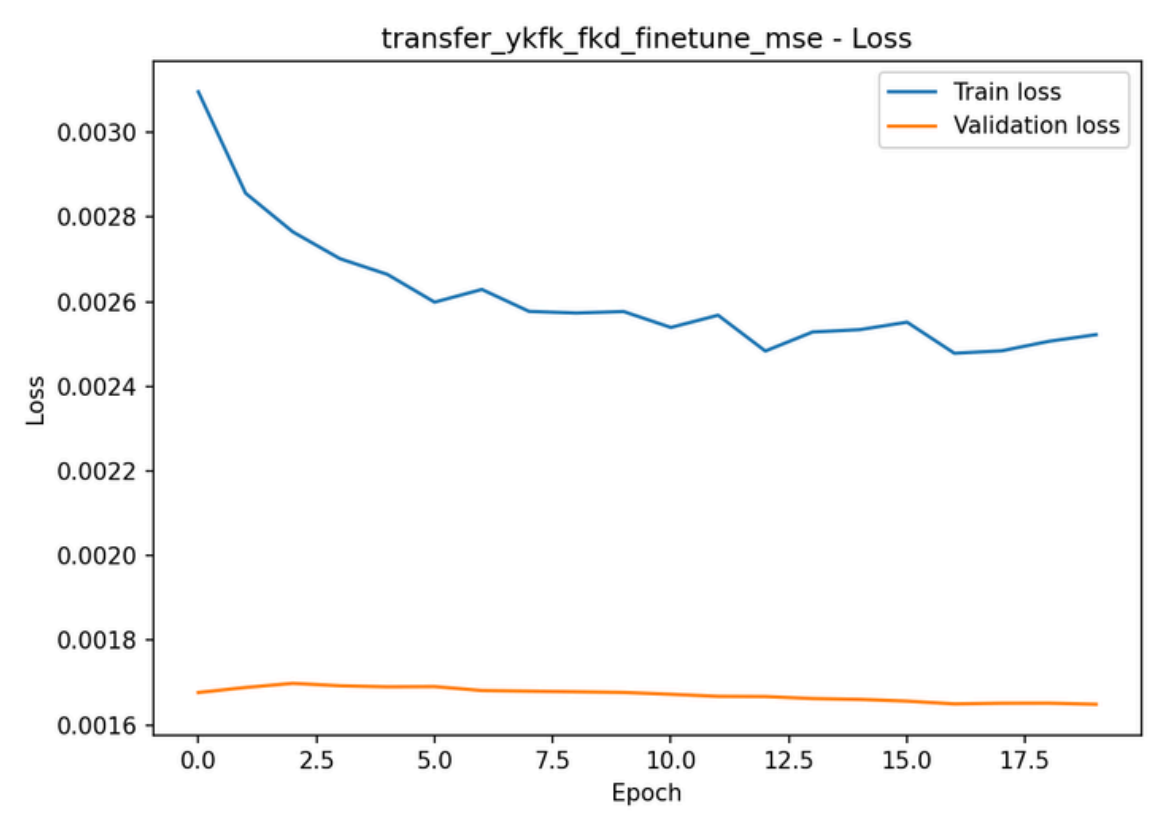
Resized and 3-Channel (150x150, 3 channels)





EXPERIMENT 4

FINAL RESULTS



experiment_name	rmse	normalized_mean_error	test_rmse	test_normalized_mean_error
mean_keypoint_baseline	3.907841	0.116925		
transfer_ykfk_fkd_finetune_mse	3.898177	0.116702	3.824378	0.119468





CONCLUSION

**The dataset
conditioned us**

**More in-depth
models/with
more parameters
do not guarantee
better results**

**Transfer learning
is useful but is
conditional on
similarity to the
dataset**





**THANK
YOU**





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